Cambridge (CIE) A Level Maths: Pure 1

Your notes

Functions

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- Inverse Functions

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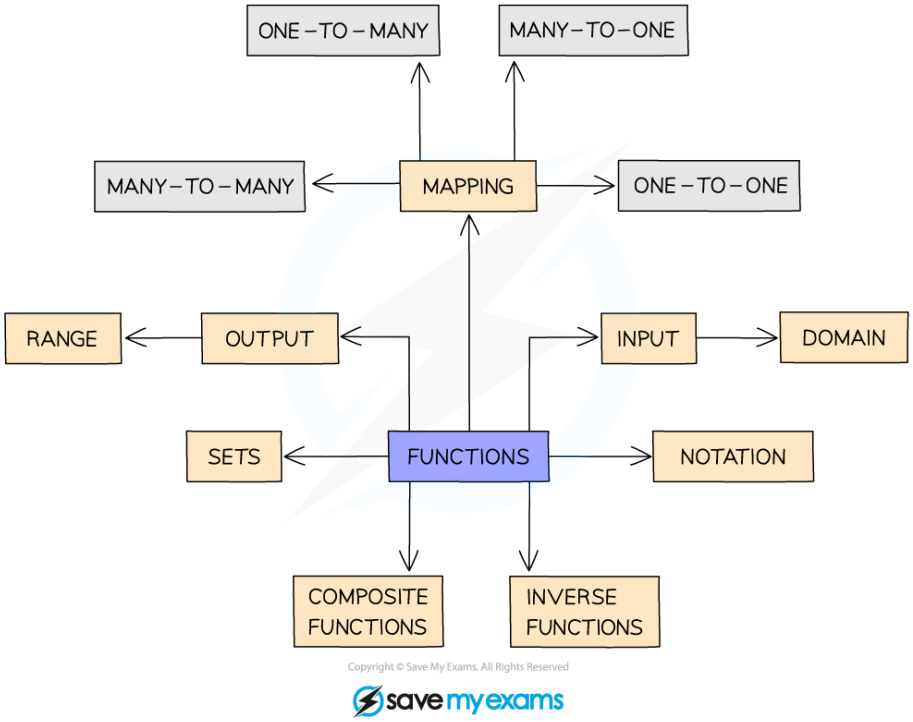
Your notes

Language of functions

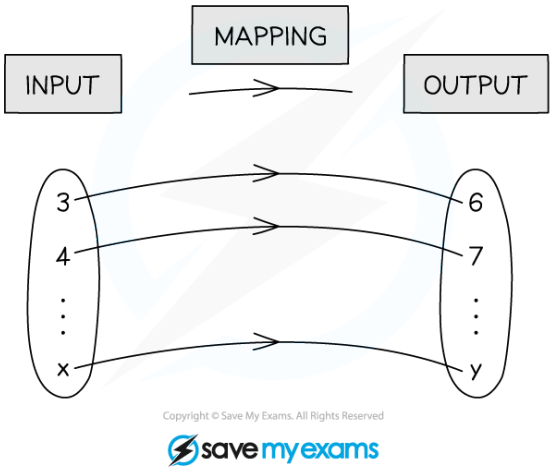
What language is used with functions?

- The language of functions has many keywords associated with it that need to be understood



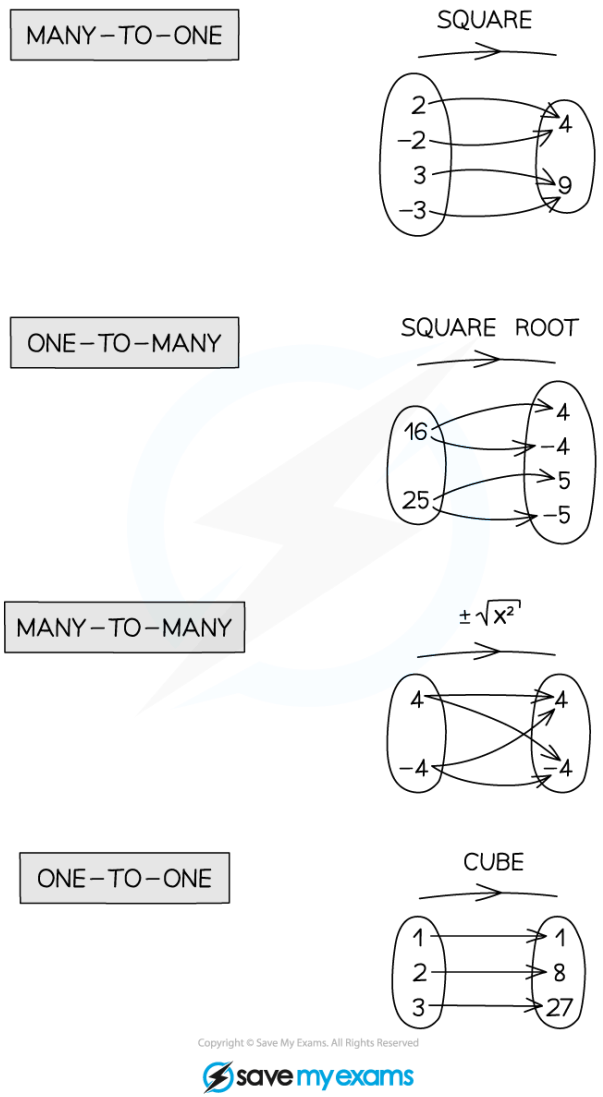
What are mappings?

- A **mapping** takes an '**input**' from one **set** of values to an '**output**' in another



Your notes

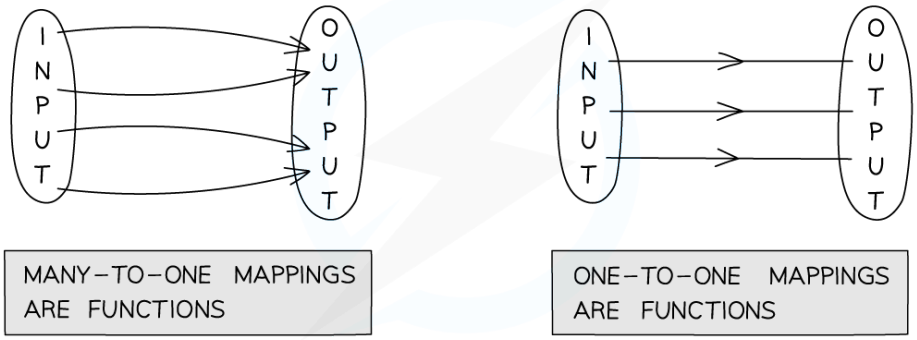
- Mappings can be
  - '**many**-to-**one**' (**many** 'input' values go to **one** 'output' value)
  - '**one**-to-**many**'
  - '**many**-to-**many**'
  - '**one**-to-**one**'



What is the difference between a mapping and a function?

- A **function** is a mapping where every 'input' value maps to a **single** 'output'
- **Many**-to-**one** and **one**-to-**one** mappings are functions
- **Mappings** which have **many possible outputs** are **not functions**

Your notes



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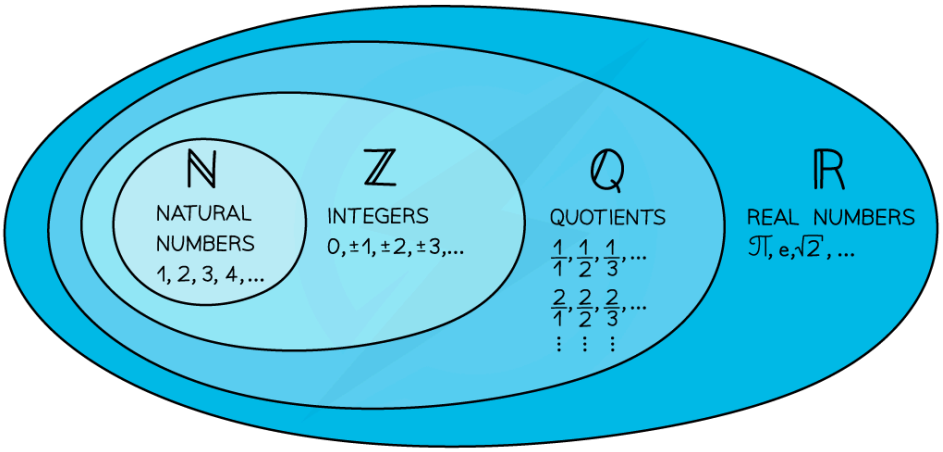
What is function notation?

- Functions are denoted by the notation **f(x)**, **g(x)**, etc
  - eg. $f(x) = x^2 - 3x + 2$
- Or the alternative notation
  - eg. $f : x \mapsto x^2 - 3x + 2$

What sets of numbers do I need to know?

- Functions often involve domains and ranges for specific sets of numbers
- All numbers can be organised into different sets $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$



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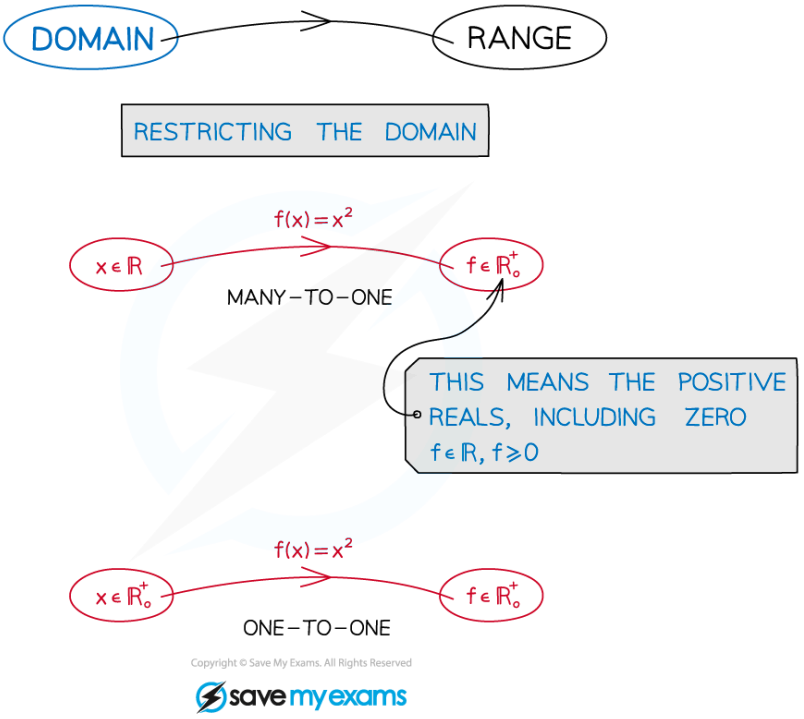
- So $\mathbb{N}$ is a **subset** of $\mathbb{Z}$ etc
- $\mathbb{Z}^-$ would be the set of negative integers only

Your notes

What is the domain of a function?

- The **domain** of a function is the set of values that are allowed to be the 'input'

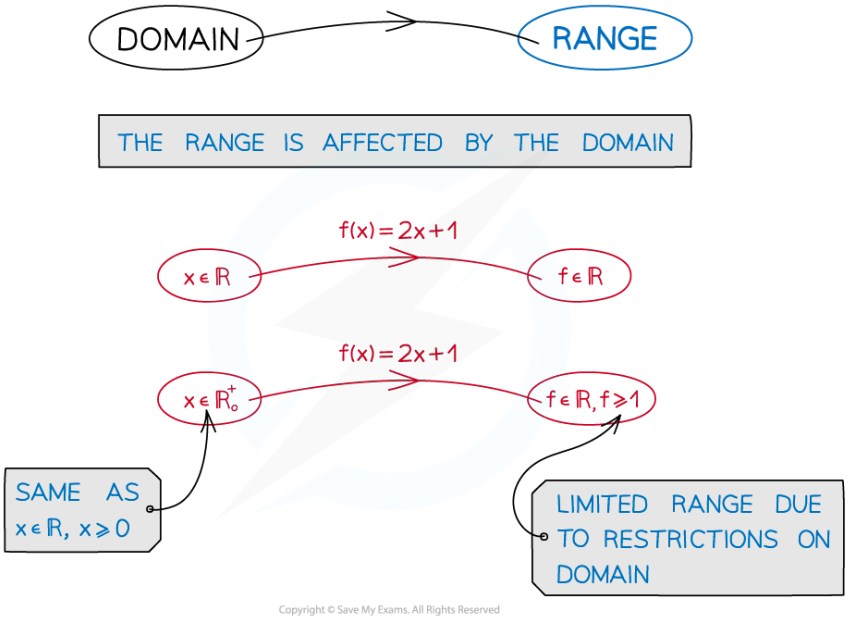
- A function is only fully defined once its domain has been stated
- **Restrictions** on a domain can turn **many**-to-**one** functions into **one**-to-**one** functions



What is the range of a function?

- The **range** of a function is the set of values of all possible 'outputs'

- The type of values in the range depend on the domain



Your notes

Worked Example

Your notes

A function is defined by $f : x \mapsto x^2 + 3x + 2, x \in \mathbb{R}, x \geq 0$

Find the range of f and sketch the graph of f against x.

POSITIVE QUADRATIC SO IT HAS A MINIMUM

$x^2 + 3x + 2 = (x + \frac{3}{2})^2 - \frac{9}{4} + 2$

$= (x + \frac{3}{2})^2 - \frac{1}{4}$

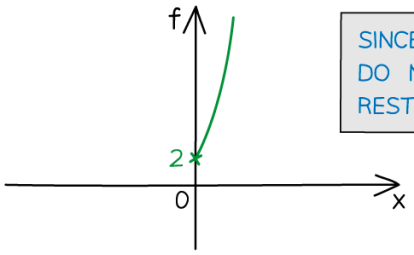
COMPLETE THE SQUARE TO FIND THE MINIMUM

$\therefore$ MINIMUM POINT IS $(-\frac{3}{2}, -\frac{1}{4})$

IT IS CRUCIAL YOU NOTICE THIS!

BUT $x \geqslant 0$ SO MINIMUM POINT OF f IS WHEN $x=0, f=2$

$\therefore$ RANGE OF f IS: $f \in \mathbb{R}, f \geqslant 2$



SINCE $x \geqslant 0$ YOU DO NOT NEED THE REST OF THE GRAPH!

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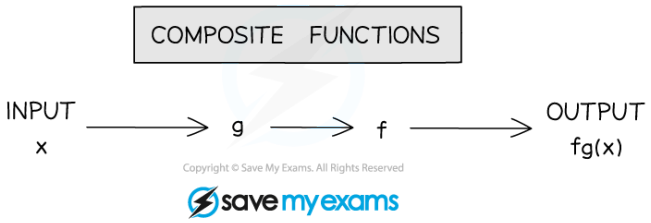
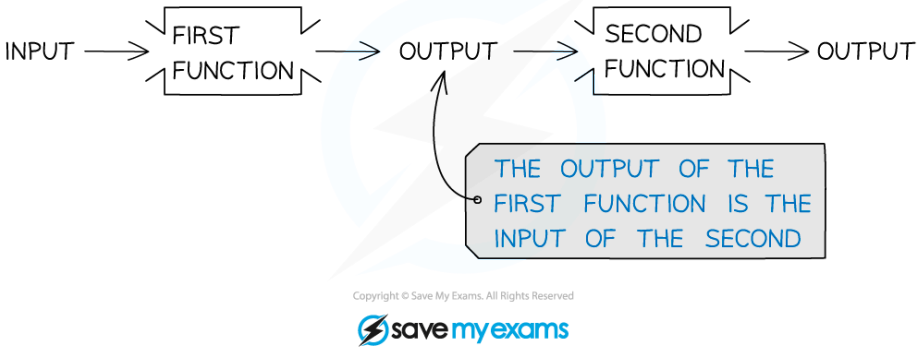
Composite functions

What is a composite function?

- A **composite function** is where one function is applied after another function

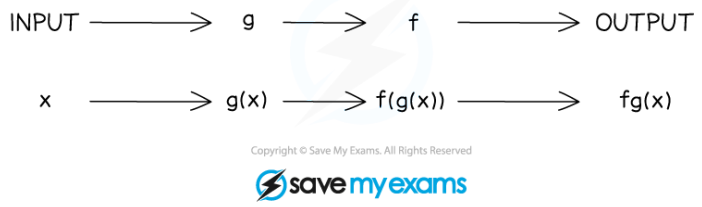


- The '**output**' of one function will be the '**input**' of the next one
- Sometimes called **function-of-a-function**
- A composite function can be denoted
  - $fg(x)$
  - $f(g(x))$
  - $f[g(x)]$
  - $(f \circ g)(x)$
  - All of these mean "f of $g(x)$"



How do I use composite functions?

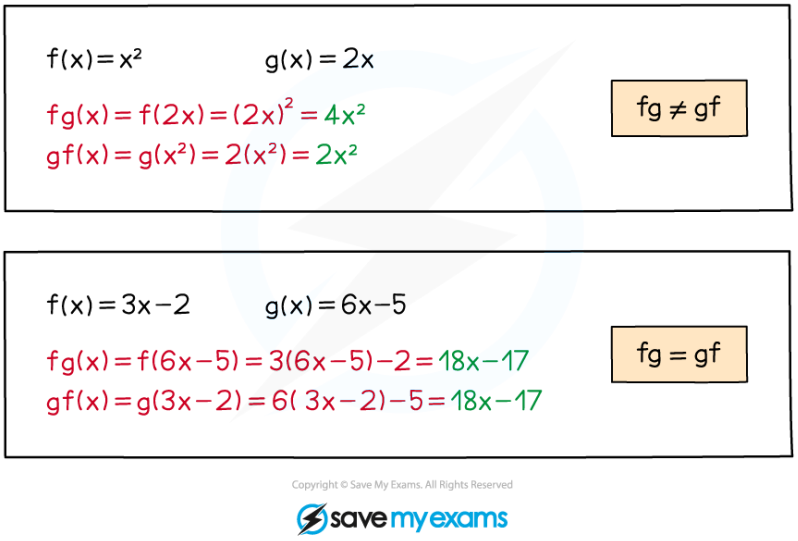
Your notes



- **Recognise** the notation
  - **fg(x)** means "**f** of **g** of **x**"

  The order matters
  - First apply *g* to *x* to get $g(x)$
  - Then apply *f* to the previous output to get $f(g(x))$
  - Always start with the function **closest to the variable**
  - $fg(x)$ is not usually equal to $gf(x)$

What are special cases of composite functions?



- **fg(x)** and **gf(x)** are generally different but can sometimes be the same
- **ff(x)** is written as **f^2(x)**
- Inverse functions **$ff^{-1}(x) = f^{-1}f(x) = x$**

Examiner Tips and Tricks

- **Domain** and **range** are important. In **fg(x)**, the 'output' (range) of **g must** be in the domain of **f(x)**, so **fg(x)** could exist, but **gf(x)** may not (or not for some values of x).

Your notes

Worked Example

Two functions $f(x)$ and $g(x)$ both have domain $x \in \mathbb{R}$

(a) Given that $f(x+3) = x^3 + 9x^2 + 27x + 26$ find $f(x)$

(b) Given that $gf(x) = 3x^3 - 3$ find $g(x)$

(c) Find $fg(x)$

(You do not need to state domain and range in this question.)

(a) $f(x+3) = x^3 + 9x^2 + 27x + 26$

USE A SUBSTITUTION AS ITS NOT OBVIOUS!

LET $y = x+3$ SO $x = y-3$

$f(y) = (y-3)^3 + 9(y-3)^2 + 27(y-3) + 26$

$= y^3 - 9y^2 + 27y - 27$

$+9y^2 - 54y + 81$

$+27y - 81$

$+26$

$= y^3 - 1$

USE BINOMIAL TO EXPAND

LINE TERMS UP NEATLY SO IT'S EASIER TO SIMPLIFY

$\therefore f(x) = x^3 - 1$

(b) $gf(x) = 3x^3 - 3 = 3(x^3 - 1)$

$= 3f(x)$

YOU COULD USE A SUBSTITUTION IF YOU DIDN'T SPOT THIS

$\therefore g(x) = 3x$

(c) $fg(x) = f(3x)$

$= (3x)^3 - 1$

g FIRST, THEN f OR "f OF g"

$\therefore fg(x) = 27x^3 - 1$

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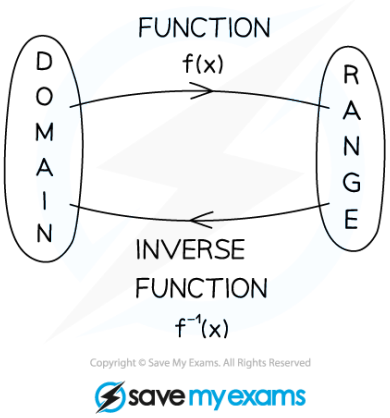
Your notes

Inverse functions

What are inverse functions?

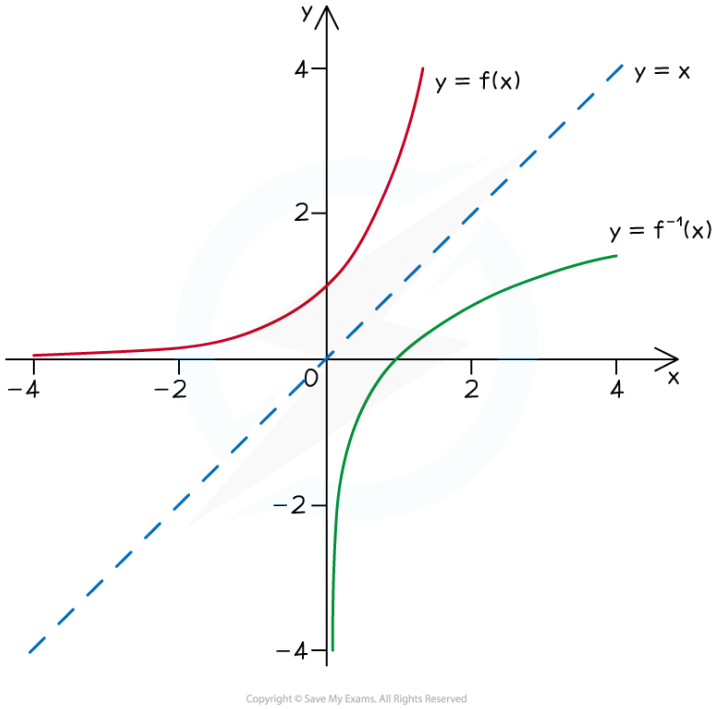
- An **inverse function** is the **opposite** to the original function
- It is denoted by $\mathbf{f^{-1}(x)}$
- An inverse only exists for **one**-to-**one** functions



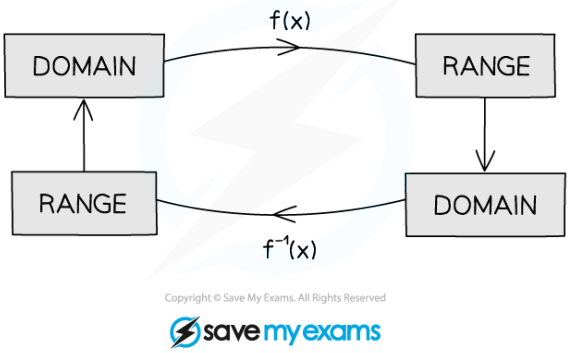
How do I sketch inverse functions?

- The graphs of a function and its inverse are reflections in the line **y = x**

Your notes



What are the domain and range of inverse functions?



- The range of a function will be the domain of its inverse function
- The domain of a function will be the range of its inverse function

How do I use algebra to find an inverse function?

- Set **y = f(x)** and make **x** the subject
- Then rewrite in function notation
- Domain is needed to fully define a function
- The range of **f** is the domain of $\mathbf{f^{-1}}$ (and vice versa)

Your notes

e.g. $f(x) = 3x^2 + 2$ $\quad x \in \mathbb{R},\ x > 0$

STEP 1: LET $y = f(x)$ $\quad y = 3x^2 + 2$

STEP 2: MAKE x THE SUBJECT

$$y - 2 = 3x^2$$

$$x^2 = \frac{y-2}{3}$$

$$x = \sqrt{\frac{y-2}{3}}$$

STEP 3: WRITE IN FUNCTION NOTATION $\quad f^{-1}(x) = \sqrt{\frac{x-2}{3}}$

WRITING IN FUNCTION NOTATION MEANS REPLACING THE y WITH AN x

STEP 4: THE DOMAIN OF f^{-1} IS THE RANGE OF f $\quad x \in \mathbb{R},\ x > 2$

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How are inverse functions and composite functions related?

- A function (**f**) followed by its inverse (**f^{-1}**) will return the input (**x**)
- **$ff^{-1}(x) = f^{-1}f(x) = x$** (for all values of x)

INPUT $\qquad$ OUTPUT

$$x \longrightarrow f \longrightarrow f^{-1} \longrightarrow x$$

$$x \longrightarrow f^{-1} \longrightarrow f \longrightarrow x$$

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Worked Example

Your notes

A function is defined as $f(x) = \sqrt{x - 2}$, for $x \in \mathbb{R}$, $x \geq 2$

(a) Write down the range of $f(x)$

(b) Find $f^{-1}(x)$ and state its domain

(c) On the same diagram sketch the graphs of $y = f(x)$ and

$$y = f^{-1}(x)$$

a) $f(2) = \sqrt{2 - 2} = 0$

$f(x) \geqslant 0$

b) $y = \sqrt{x - 2}$ ← STEP 1: $y = f(x)$

$y^2 = x - 2$

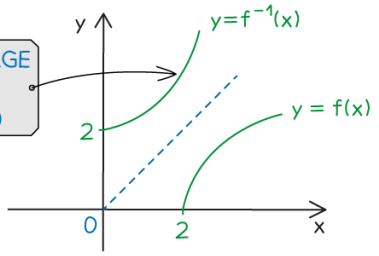
$x = y^2 + 2$ ← STEP 2: MAKE x THE SUBJECT

$\therefore f^{-1}(x) = x^2 + 2$ ← STEP 3: FUNCTION NOTATION

$x \in \mathbb{R}, x \geqslant 0$ ← STEP 4: DOMAIN OF f^{-1} IS RANGE OF $f(x)$

c)

USE MIRROR IMAGE IN LINE $y=x$ TO SKETCH $y=f^{-1}(x)$



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